

Supplementary material for the manuscript entitled “Prediction of GO Terms Based on Partitioning PPI Networks into Highly Connected Components”, ID: TCBB-2022-11-0938

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This document contains supplementary material supporting the manuscript titled “*Prediction of GO Terms Based on Partitioning PPI Networks into Highly Connected Components*”. The additional content provided herein includes extended experimental results, supplementary tables and figures, detailed examples of the GO term prediction process, as well as further elaboration on the methodology and dataset descriptions.

1 Biological interpretation of metabolite HCD components

As it is stated in main paper, some of the HCD components obtained by partitioning given metabolic networks are analyzed from the biological point of view. More precisely, the process of fatty acid synthesis is recognized in HCD components of all four metabolic networks. This process consists of three consecutive phases: activation, elongation and termination [1, 2]. In each of the phases, some of the intermediates are repeated, so in some organisms the metabolites participating in these phases are denser, whereas in others they are less frequently related. Based on metabolic networks, a network corresponding to one of the phases of fatty acid synthesis is obtained as an HCD component for all four organisms. Figure 1 shows one of the highly related components of the *Saccharomyces cerevisiae* organism. This HCD component represents the termination phase of the fatty acid synthesis process. On the other hand, the HCD component of the metabolic network *Staphylococcus aureus* is a structure which represents the first two phases of the fatty acid synthesis process, i.e., the phases of activation and elongation (Figure 2). Unlike the organism *Staphylococcus aureus*, where the intermediates of the first two phases of the fatty acid synthesis process are connected into one HCD component, the activation phase for the organism *Tuberculosis – Mycobacterium tuberculosis* is separated into a separate component (Figure 3). For the organism *Escherichia coli*, one of the HCD components represents the elongation phase of the fatty acid synthesis process (Figure 4).

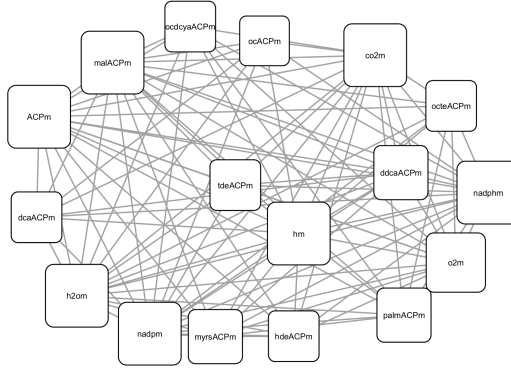


Figure 1: The termination mechanism in fatty acid synthesis in *Saccharomyces cerevisiae* network

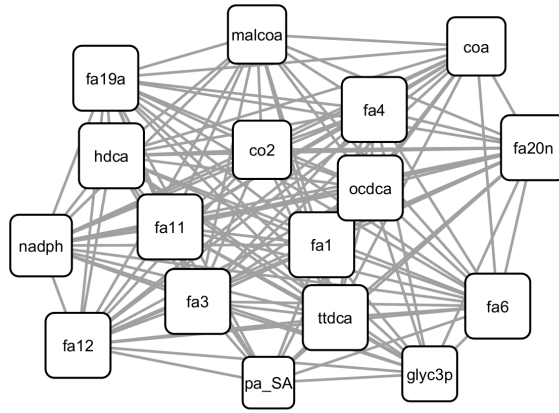


Figure 2: Activation and elongation phases in fatty acid synthesis in *Staphylococcus aureus* network

These findings are in the line with conclusions from paper [3]. More precisely, in paper [3] a network is partitioned into subgraphs, called k -plexes where the degree of each vertex within the k -plex with n vertices is at least $n - k$. In the same study is demonstrated that relaxing the partitioning requirements by varying k can lead to more descriptive and biologically relevant results. Since, the HCD components can be considered as k -plexes with $k = n/2 - 1$, they represent more relaxed partitioning scheme which can lead to more descriptive and biologically relevant results. It is important to mention that in the paper [3] solely to the metabolic network of *Saccharomyces cerevisiae* (yeast) was considered, while in our study, we extend the analysis to include multiple organisms. It is important to note that the k -plex partitioning has not been applied to PPI networks, which prevents us from making a direct comparison with methods of that type.

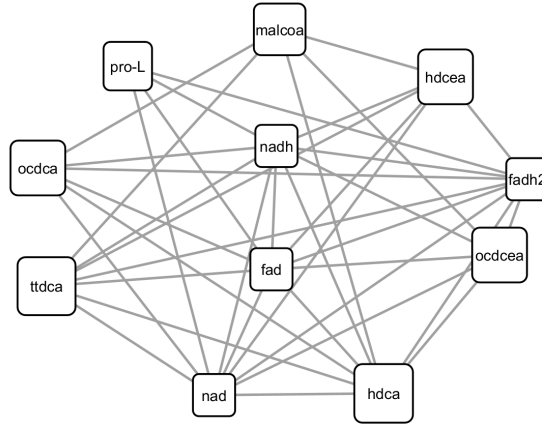


Figure 3: The activation phase in fatty acid synthesis in *Tuberculosis* –*Mycobacterium tuberculosis* network

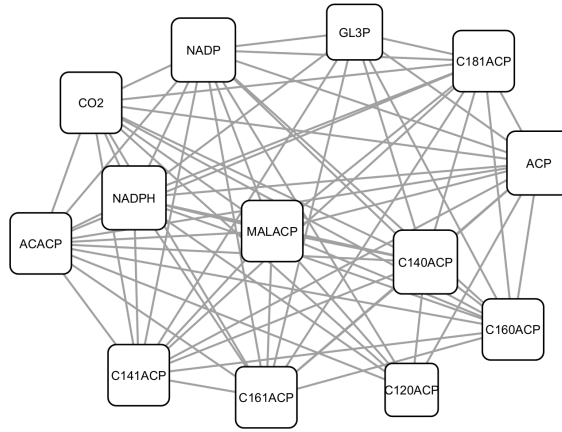


Figure 4: The elongation phase in fatty acid synthesis in *Escherichia coli* network

2 Parameters of VNS and GOTerm prediction methods

In this section, we provide an overview of the key parameters used in the algorithms. A detailed description of each parameter, along with its value is shown in Tables 1 and 2, is provided to ensure clarity and to allow for reproducibility of the experiments.

Table 1: VNS Parameters

Parameter	Description	Value
k_{min}	Minimum size of neighborhood in Shaking	1
k_{max}	Maximum size of neighborhood in Shaking	20
$iter_{total}$	Total iterations performed in VNS	10000
$iter_{withoutImpr}$	Maximum iterations without improvement before stopping	2000
$prob$	Probability switching to a new solution when objective functions are equal	0.5
max_time	Maximum time allowed for the algorithm to run	4 hours

Table 2: Parameters for GO Prediction Method

Parameter	Value
Number of repetitions	100
Number of different values for Score1	10
Number of different values for Score2	10
Range for Score1	0.001 – 0.05
Step for Score1	0.0054
Range for Score2	0.001 – 0.999
Step for Score2	0.11
Metric used for evaluation	F1 measure

3 Stability and Sensitivity Analysis

To provide a comprehensive quantitative characterization of the proposed VNS-based method, we evaluate its computational stability, sensitivity across multiple runs, and structural consistency of the obtained solutions on the benchmark datasets considered in the main manuscript. The analysis covers both optimization robustness and the reproducibility of the resulting partitions. A complementary evaluation on datasets from additional application domains is provided in Section 5, Table 6.

Stability is evaluated by calculating the **average gap**:

$$\text{AvgGap} = \frac{1}{R} \sum_{i=1}^R \frac{f_i - f_{\text{best}}}{f_{\text{best}}}$$

and the corresponding **standard gap deviation**:

$$\text{StdGap} = \sqrt{\frac{1}{R} \sum_{i=1}^R \left(\frac{f_i - f_{\text{best}}}{f_{\text{best}}} - \text{AvgGap} \right)^2}$$

across all runs ($R = 10$) for each network instance. Here, f_i denotes the objective value obtained in the i -th run, and f_{best} is the best objective value observed over all runs.

Structural sensibility of the obtained partitions is assessed using the **Adjusted Rand Index (ARI)**. For each pair of runs, ARI measures the similarity between the two node partitions by comparing all pairs of nodes and counting how consistently they are assigned to the same or to different components in both partitions. The index is corrected for chance, so that a value of 1 indicates identical partitions, while a value close to 0 corresponds to agreement expected at random. In our experiments, ARI values are computed for all pairs of runs of the same instance, and the reported value represents the average similarity across runs.

All results for the benchmark datasets considered in the main manuscript are summarized in Table 3.

Table 3: Stability and sensitivity statistics of the VNS algorithm

instances	AvgGap	StdGap	Mean_ARI	Std_ARI
At-all	0.001449	0.000769	0.8863	0.0171
AT-phys	0.001284	0.00059	0.8822	0.0198
Ce-all	0.001807	0.000783	0.7483	0.0588
Ce-phys	0.000405	0.000265	0.7483	0.0588
Mm-all	0.000947	0.000525	0.8173	0.0263
Mm-phys	0.000372	0.000361	0.8628	0.0202
Sp-all	0.002626	0.001638	0.649	0.0387
Sp-phys	0.00068	0.000632	0.9187	0.023
Ec	0.002275	0.002638	0.9083	0.0518
Sa	0.000584	0.000485	0.9888	0.0032
Tu	0.001954	0.001465	0.8665	0.0465
Sc	0.003324	0.002224	0.8412	0.0447
gs_50	0	0	1	0
gs_80	0	0	0.9824	0.0188
gs_100	0	0	0.99	0.0093
gs_150	0	0	1	0
gs_200	0	0	1	0
gs_500	0	0	1	0
gs_1000	0	0	1	0
gs_2000	0	0	1	0
gs_5000	0	0	1	0
gs_10000	0	0	1	0
gd_50	0.005028	0.001676	0.7453	0.3293
gd_80	0.000251	0.000754	0.9574	0.0852
gd_100	0	0	0.9779	0.0442
gd_150	0	0	1	0
gd_200	0	0	1	0
gd_500	0	0	1	0
gd_1000	0	0	1	0
gd_2000	0	0	1	0
gd_5000	0	0	0.9982	0.0027
gd_10000	0	0	1	0

4 Complexity of VNS algorithm

Complexity analysis is performed only for the local search procedure, since it represents the most computationally expensive part of the overall algorithm.

A single iteration of the local search algorithm may, in the worst case, be executed for each node $i \in V$. For each such node, the algorithm attempts to move i from its current component p ($i \in V_p$) to any other existing component q , as well as to a new singleton component.

Let $\Delta(G)$ denote the maximum degree of a node in G . For each hypothetical move, a partial evaluation of the objective function is performed. This evaluation runs, in the worst case, in time

$$O(\max(|V_p|, |V_q|, \Delta(G))).$$

The term $|V_p|$ arises because the original component must be examined and the correctness values of its nodes updated. The term $|V_q|$ corresponds to updating the correctness values in the destination component. The term $\Delta(G)$ appears because all neighbors of node i must be inspected to determine whether any of them belong to the target component q ; if none do, a new component is formed.

A naive upper bound on the time complexity of one local search iteration is therefore

$$O(|V| \cdot |V| \cdot |V|) = O(|V|^3),$$

which corresponds to the case where there are many components of large cardinality. Although this bound is formally correct as an upper bound, it does not represent a realistic scenario, since it is not possible to simultaneously have many components of large size.

We therefore consider several special cases.

Case 1: A single component. If all nodes belong to a single component, the outer loop iterates over $|V|$ nodes, and for each node only one move is considered, namely moving it to its own singleton component. In this case, the partial evaluation of the objective function runs in time

$$O(\max(|V|, 0, \Delta(G))) = O(|V|),$$

and the total complexity of one iteration of local search is $O(|V|^2)$.

Case 2: All components are singletons. If every node forms its own component, the outer loop runs for $|V|$ iterations, while the inner loop considers $|V| - 1$ possible destination components. The partial evaluation then runs in time $O(\Delta(G))$, yielding an overall complexity of

$$O(|V|^2 \cdot \Delta(G)).$$

Intermediate cases. All intermediate cases are cheaper in terms of time complexity than $O(|V|^2 \cdot \Delta(G))$. For example, suppose there are m components of approximately equal size $|V|/m$, where m is a constant smaller than $|V|$ and larger than 1. The outer loop iterates over $|V|$ nodes, the inner loop over m components, and the partial objective evaluation runs in time

$$O\left(\max\left(\frac{|V|}{m}, \Delta(G)\right)\right).$$

If $\Delta(G) \geq |V|/m$, the total complexity is $O(|V| \cdot m \cdot \Delta(G)) \leq O(|V|^2 \cdot \Delta(G))$. Otherwise, if $|V|/m > \Delta(G)$, the complexity becomes

$$O\left(|V| \cdot m \cdot \frac{|V|}{m}\right) = O(|V|^2),$$

which is again upper-bounded by $O(|V|^2 \cdot \Delta(G))$.

We therefore conclude that the time complexity of a single iteration of the local search algorithm is

$$O(|V|^2 \cdot \Delta(G)).$$

The total number of local search iterations cannot be determined in advance, since the procedure is executed repeatedly as long as improvements are found. Therefore, we estimate the number of local search iterations empirically by performing a single run on artificial random instances with up to 1000 nodes. The obtained results are reported in Table 4.

The first four columns have been explained previously. Regarding the remaining columns, *VNS iter.* denotes the number of iterations of the main VNS loop, which is unknown in advance but is upper bounded by a user-defined control parameter; *LS iter.* represents the total number of local search iterations, which is also unknown beforehand; *Iter. ratio* gives the ratio between these two values; *Iter. cost* corresponds to an upper bound on the cost of a single local search iteration; and *Cost* provides an estimate of the total computational cost of all local search iterations, computed as the product of *LS iter.* and *Iter. cost*. The last column reports the actual execution time.

As shown below the table, the estimated total cost is highly correlated with the realized execution time ($\rho \approx 0.98$), which confirms the validity of the proposed upper bound on the complexity of a single local search iteration. Concerning *LS iter.*, as already noted, it cannot be directly bounded in advance, since it depends on the specific problem instance and its internal structure. However, inspection of the *Iter. ratio* column reveals that higher graph density leads to larger ratios. In particular, for sparse instances, the number of local search iterations is up to approximately 1.3 times the number of VNS iterations, whereas for dense instances this factor increases to about 4.5.

Nevertheless, the total number of local search iterations exhibits an approximately linear dependence on the maximum number of VNS iterations, which is a user-specified control parameter. Consequently, the overall time complexity can be expressed as the product of the maximum number of VNS iterations and $O(|V|^2 \cdot \Delta(G))$.

In addition to the complexity of the VNS-based decomposition, it is important to account for the cost of the extension phase (Phase 2 of the HCDAM), in which singleton nodes are assigned to the previously identified HCD components.

The assignment procedure is implemented by traversing the adjacency list of each singleton node and aggregating the interaction scores with nodes belonging to different components. For a node of degree d , this requires $O(d)$ time, assuming constant-time access to component labels. Consequently, the total cost of assigning all singleton nodes can be expressed as $O(\sum_{v \in S} d(v))$, where S denotes the set of singleton nodes, which is bounded by $O(|E|)$.

Therefore, the extension phase is linear in the size of the input graph and represents an efficient and scalable post-processing step.

Table 4: Empirical analysis of estimated and realized complexity for artificial instances with up to 1000 nodes.

Instance	n	m	$density$	$\Delta(G)$	VNS iter.	LS iter.	Iter. ratio	Iter. cost	Cost	$t[s]$
gs_50	50	102	0.083	9	50000	52699	1.05	22500	1.19×10^9	6.43
gs_80	80	149	0.047	9	50011	61156	1.22	57600	3.52×10^9	7.65
gs_100	100	312	0.063	15	50127	64350	1.28	150000	9.65×10^9	11.67
gs_150	150	0.078	875	21	50003	57202	1.14	472500	2.70×10^{10}	19.93
gs_200	200	0.071	1410	23	50006	50006	1.00	920000	4.60×10^{10}	25.33
gs_500	500	4035	0.032	26	50003	50003	1.00	6500000	3.25×10^{11}	103.84
gs_1000	1000	8409	0.017	27	50001	50002	1.00	27000000	1.35×10^{12}	343.88
gd_50	50	244	0.199	16	57696	150595	2.61	40000	6.02×10^9	14.32
gd_80	80	527	0.167	21	54946	219831	4.00	134400	2.95×10^{10}	32.91
gd_100	100	858	0.173	28	55540	250034	4.50	280000	7.00×10^{10}	60.09
gd_150	150	2149	0.192	47	50070	100477	2.01	1057500	1.06×10^{11}	54.34
gd_200	200	3818	0.192	59	50247	90951	1.81	2360000	2.15×10^{11}	72.67
gd_500	500	18381	0.147	100	50599	85956	1.70	25000000	2.15×10^{12}	298.79
gd_1000	1000	63406	0.127	163	50469	102441	2.03	163000000	1.67×10^{13}	1219.77
					$\rho(density, Iter.ratio) \approx 0.69$			$\rho(Cost, t[s]) \approx 0.98$		

5 Extension of the Proposed Methodology to Other Network Domains

This section illustrates the general applicability of the procedure introduced in the main study beyond the original domain of PPI networks. The underlying methodology follows three conceptual steps:

- Identification of highly connected components: The VNS heuristic is used to detect dense subgraphs that represent strongly interconnected units within the network, according to the introduced optimization procedure based on the HCD problem.
- Extension of components with singleton nodes: Isolated or peripheral nodes are assigned to existing dense components using a strategy that maximizes the structural cohesion of the extended component. In the PPI context, this assignment

Table 5: Summary of VNS results on 10 collaborative and 5 economic networks

Network	n	m	density	Best	Avg	Reduct.	AvgVNS	t[s]	Larg.
sandi-auths	86	124	0.03393	75	75	38	37	6.5623	5
netscience	379	914	0.01276	451	452.4	37	415.4	167.0402	9
CSphd	1882	1740	0.00098	1731	1731	1720	11	1.192	3
GrQc	4158	13422	0.00155	6087	6092.1	1252	4840.1	14400.14	46
Erdos992	5094	7515	0.00058	6986	6989.3	5210	1779.3	2443.525	11
HepPh	11204	117619	0.00187	50679	51689.6	2369	49320.6	14402.75	229
AstroPh	17903	196972	0.00123	129085	129539	2050	127489	14406.76	75
CondMat	21363	91286	0.0004	61255	61349.8	3447	57902.8	14409.95	30
cit-HepTh	22908	2444798	0.00932	1901104	1907186	453	1906733	14419.57	539
cit-HepPh	28093	3148447	0.00798	2480859	2491732	267	2491465	14440.73	421
wml	258	2389	0.07205	1017	1025.1	394	631.1	155.4149	60
mahindas	1258	7513	0.0095	5409	5411.5	3508	1903.5	1352.315	79
orani678	2529	86768	0.02714	40849	41252.2	8762	32490.2	14424.78	367
psmigr1	3140	410781	0.08335	300749	301431.8	6	301425.8	14402.16	485
poli-large	15575	17468	0.00014	16785	16786.7	15103	1683.7	5496.305	5

was guided by the average combined score (ACS) with component members; in other domains, an analogous structural criterion is used.

- Domain-specific interpretation: Once extended, components are analyzed according to the context of the network.

In order to assess the applicability of the proposed method beyond its original setting, we extend the analysis to two additional classes of networks, namely collaborative and economic networks. For this purpose, a dataset consisting of 10 collaborative and 5 economic networks was prepared based on publicly available network datasets from the Network Repository [4]. The selected instances were chosen to be consistent with the experimental setting considered in the main study, in particular with respect to network size, with the number of nodes ranging from several hundreds to approximately 28000.

Table 5 summarizes the performance of the proposed VNS-based approach across all considered instances. The first ten rows correspond to collaborative networks, while the remaining five rows represent economic networks. For each network, the table reports the number of nodes and edges, along with the corresponding graph density. The columns *Best* and *Avg* denote the best and average objective values obtained over multiple runs, respectively. To assess the effectiveness of the search process, we report the average number of removed edges in the reduction phase (*Reduct.*), as well as the lowest and average number of edges removed by the VNS procedure (*AvgVNS*) across 10 independent VNS runs. Finally, *t[s]* denotes the average computational time, while *Larg.* reports the size of the largest extracted component, reflecting the ability of the method to preserve substantial cohesive structures within the network.

The robustness and stability of the proposed method on collaborative and economic networks are reported in Table 6. The table is organized on the same way as Table 3.

The main observations can be summarized as follows:

- High solution quality: The algorithm achieves low average optimality gaps across all instances, often below 1%, and in several cases equal to zero (e.g., *sandi-auths*, *CSphd*), indicating convergence to high-quality solutions.

Table 6: Stability and sensitivity statistics of the VNS algorithm on collaborative and economic networks

instances	AvgGap	StdGap	Mean_ARI	Std_ARI
sandi-auths	0	0	0.8932	0.051
netscience	0.3104	0.203218	0.8591	0.0398
CSphd	0	0	0.7624	0.1646
GrQc	0.0838	0.074001	0.8523	0.011
Erdos992	0.0472	0.061084	0.8889	0.0158
HepPh	1.9941	0.857261	0.7509	0.009
AstroPh	0.3517	0.155335	0.7661	0.0068
CondMat	0.1548	0.096517	0.7889	0.0055
cit-HepTh	0.3199	0.232675	0.7419	0.0178
cit-HepPh	0.4383	0.197023	0.7471	0.0069
wm1	0.7965	1.216613	0.5813	0.1339
mahindas	0.0462	0.060333	0.9519	0.0139
orani678	0.987	0.885391	0.8885	0.0202
psmigr1	0.227	0.151808	0.8512	0.0097
poli-large	0.0101	0.01193	0.8646	0.0121

- **Preservation of cohesive structures:** Despite enforcing strong degree constraints, the largest connected components remain nontrivial in size across all networks, indicating that the method preserves meaningful dense substructures.
- **Scalability:** The method handles networks ranging from tens to tens of thousands of nodes, maintaining consistent performance, albeit with increased computational time for the largest instances.
- **Robustness and stability:** The standard deviation of the gap remains low for most networks, showing that the method produces stable solutions across multiple runs, even for large-scale instances.
- **Sensitivity:** In most cases, the mean ARI values are consistently high (typically above 0.75), with very low standard deviations, indicating that the structure of the obtained components is largely insensitive to random initialization and search variability. Notably, several instances such as *econ-mahindas* and *ca-Erdos992* achieve mean ARI values close to or above 0.88, with minimal variance, reflecting highly stable decomposition behavior. A slightly higher variability is observed in a small number of cases (e.g., *ca-CSphd* and *econ-wm1*), which can be attributed to the smaller or structurally more heterogeneous nature of these networks.

5.1 Further analysis on Collaborative Networks

From a domain-specific perspective, the dense components detected in collaborative networks can be naturally interpreted as tightly connected research teams. Authors within these components maintain a high proportion of mutual collaborations, indicating sustained and intensive joint research activity. This interpretation aligns with the established understanding of co-authorship networks, where densely connected subgraphs are commonly associated with stable and productive scientific collaborations.

Following the extraction of dense components, a subset of nodes remains outside these highly cohesive substructures. In the context of collaborative networks, such nodes can be interpreted as peripheral or weakly integrated researchers. In many cases, these authors act as occasional collaborators, interdisciplinary contributors, or early-stage researchers whose publication records have not yet led to strong integration within a specific research group.

To refine the structural organization of the network, singleton nodes are assigned to the previously identified dense components according to a natural structural criterion: each singleton is associated with the component with which it shares the largest number of adjacent nodes.

To assess the effect of singleton assignment, we analyzed the components obtained from the VNS solution with the lowest objective function value across 10 runs. Table 7 summarizes the main structural characteristics of the obtained components before (B) and after (A) extension, including the number of nontrivial components ($\#C$), number of nodes and edges included in all nontrivial components for both phases, as well as their average internal density. The rightmost column reports the density ratio between the average density after expansion and the initial network density.

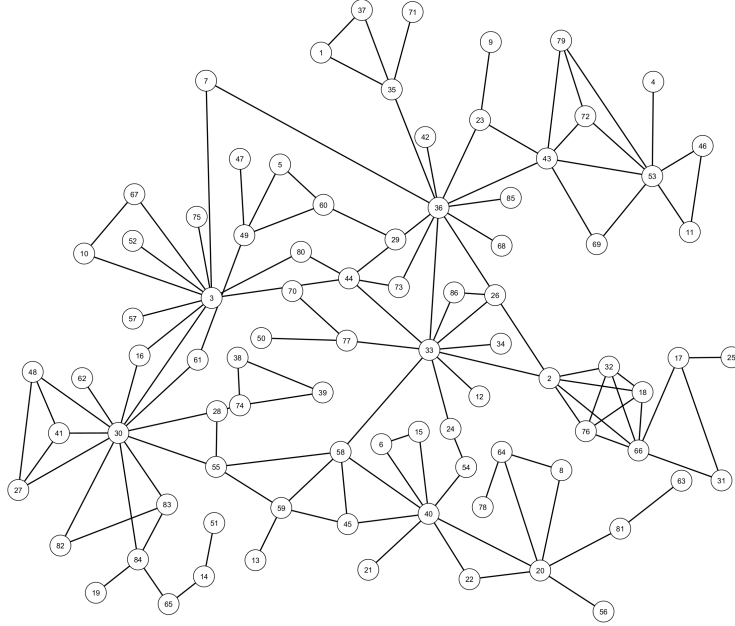


Figure 5: Original structure of the *sandi-auths* collaboration network prior to processing.

As expected, the inclusion of singleton nodes leads to a substantial increase in both the number of nodes and edges across all instances. The expansion is accompanied by a decrease in the average density of components, with values typically dropping from near-clique levels (close to 1) to a range between approximately 0.10 and 0.95 after

Table 7: Overall statistics of collaboration networks after singleton expansion.

Network	#C	#N (B)	#N (A)	#E (B)	#E (A)	A.Den. (B)	A.Den. (A)	Den.R.
sandi-auths	13	42	77	49	88	1	0.5365	15.8
netscience	56	243	372	463	685	0.99	0.6925	54.3
CSphd	3	9	38	9	43	1	0.2583	262.8
GrQc	596	2466	3939	7335	9312	0.9916	0.6547	421.6
Erdos992	123	414	3914	529	4319	0.9850	0.0980	169.2
HepPh	1481	7341	10979	66940	72077	0.9935	0.7128	380.3
AstroPh	2478	13162	17704	67887	76172	0.9887	0.7962	647.8
CondMat	3533	14653	20941	30031	39212	0.9919	0.7497	1874.1
cit-HepTh	2463	20143	22753	543694	562167	0.9787	0.9301	99.8
cit-HepPh	2945	25022	28044	667588	688561	0.9810	0.9490	118.9

extension. Despite this reduction, the resulting densities remain significantly higher than the overall network density, as reflected by the large density ratio values, which range from approximately 15 to 1874. This confirms that the expanded components preserve a high level of internal cohesion while substantially extending their coverage.

Notably, the extent of density reduction varies across networks, reflecting differences in their underlying collaboration patterns. In several cases (e.g., *cit-HepTh*, *cit-HepPh*), the density remains very high even after expansion, since the initial components already cover a large portion of the network. Even though several thousand nodes and tens of thousands of edges are added, their effect on the overall density remains marginal due to the already extensive and cohesive core structure. In contrast, networks such as *Erdos992* exhibit a more pronounced decrease in density, reflecting their inherently heterogeneous collaboration patterns, where a large number of peripheral authors are connected through relatively sparse and irregular co-authorship links.

Table 8: Detailed quantitative data overview obtained on Sandi-auths collaborative network, before and after extension

Comp.	Nodes (B)	Nodes (A)	Edges (B)	Edges (A)	Density (B)	Density (A)
1	3	7	3	7	1	0.3333
2	3	9	3	9	1	0.25
3	3	5	3	5	1	0.5
4	4	11	6	15	1	0.2727
5	3	5	3	6	1	0.6
6	3	7	3	7	1	0.3333
7	5	7	10	13	1	0.619
8	3	3	3	3	1	1
9	3	5	3	5	1	0.5
10	3	3	3	3	1	1
11	3	4	3	4	1	0.6667
12	3	5	3	5	1	0.5
13	3	6	3	6	1	0.4

As a representative example, a more detailed analysis of the singleton expansion process is provided for the *sandi-auths* collaboration network. The structural evolution of the network across the three stages is illustrated in Figures 5–7, which depict the original network, the decomposition obtained via the proposed VNS-based procedure, and the final structure after singleton expansion, respectively. Initially, the network exhibits a sparse topology without clearly distinguishable community structure. After applying the VNS-based decomposition, it is partitioned into a set of highly connected compo-

nents with dense internal organization, corresponding to tightly integrated collaboration groups. Finally, singleton nodes are incorporated into these components, leading to larger but still structurally cohesive subgraphs that better reflect the underlying collaboration patterns.

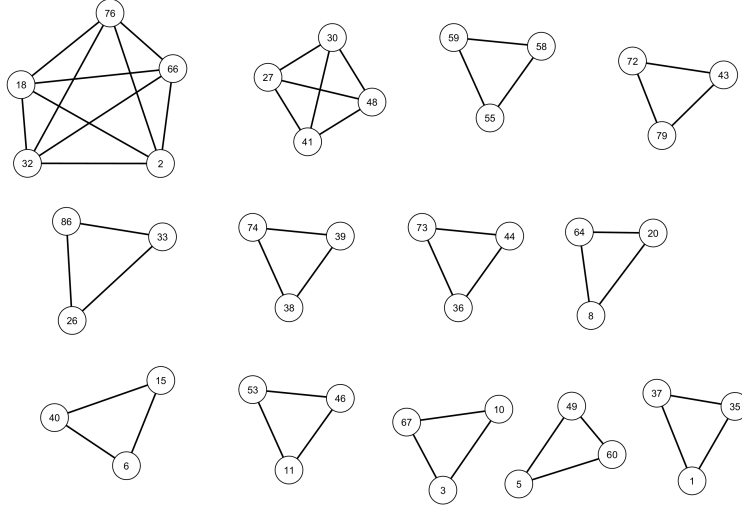


Figure 6: Network decomposition obtained after applying the VNS-based procedure, showing extracted dense components.

Table 8 complements this visual analysis by quantifying the structural effects at the level of individual components. As expected, all initially extracted components exhibit clique-like structures, reflected by unit density prior to expansion, which is a direct consequence of the VNS-based decomposition that isolates highly cohesive subgraphs. After singleton inclusion, a consistent increase in component size is observed, accompanied by a moderate decrease in density; however, the resulting values remain relatively high across all components.

The results indicate that singleton assignment leads to a controlled relaxation of highly connected components into larger but still dense subgraphs. This confirms that the proposed extension mechanism preserves the internal structural integrity of components while incorporating peripheral nodes in a meaningful way.

All data related to additional collaborative and economic networks are publicly available in the GitHub repository¹.

¹<https://github.com/milanagrbic/HCDAM4GOTermPrediction>

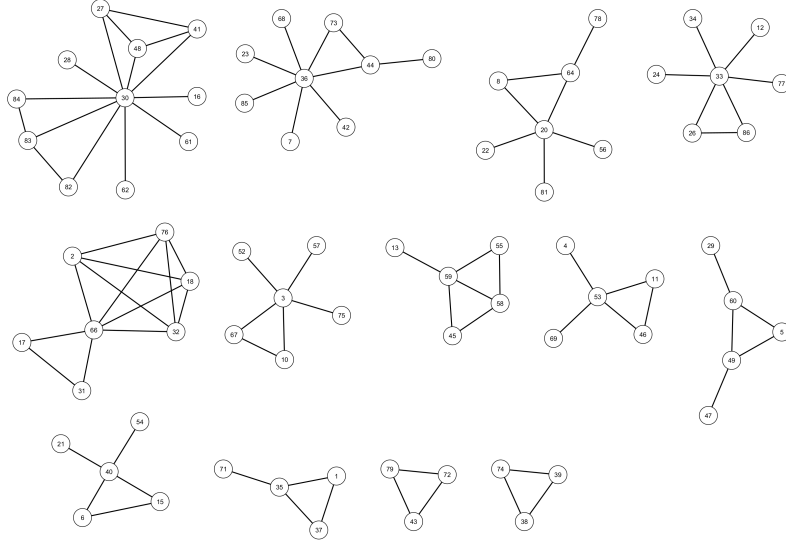


Figure 7: Final structure after singleton expansion, where peripheral nodes are assigned to dense components.

5.2 Further analysis on Economic Networks

In the case of economic networks, we consider two variants of the singleton assignment strategy. In the first variant, edge weights are ignored, and the assignment rule follows the same principle as in collaboration networks, where each singleton is attached to the component with which it shares the largest number of adjacent nodes. In the second variant, edge weights are explicitly taken into account, and singleton nodes are assigned to the component with which they exhibit the strongest interaction, measured in terms of aggregated edge weights. In the weighted setting, it is important to note that edge weights in economic networks have different interpretations depending on the underlying dataset. For instance, in the *psmigr1* network, edge weights represent the number of individuals migrating between locations, reflecting the intensity of population flows. In contrast, in the *mahindas* network, weights correspond to economic interactions between sectors, capturing the magnitude of interdependencies within the system. Furthermore, some economic networks contain negative edge weights, which typically indicate inverse or inhibitory relationships between entities. Since the focus of this study is on the strength of interaction rather than its direction or sign, absolute values of edge weights are considered in the analysis.

From a domain-specific perspective, the highly connected components identified in economic networks can be interpreted as groups of strongly interdependent entities, such as sectors, regions, or agents, characterized by intensive mutual interactions. Nodes that remain outside these components correspond to entities with weaker or more dispersed connections, either lacking strong integration within any single dense component or maintaining only limited links across multiple groups. In both the unweighted and

Table 9: Overall statistics of economic networks after singleton expansion.

Network	#C	#N (B)	#N (Anw)	#N (Aw)	#E (B)	#E (Anw)	#E (Aw)	A.Den. (B)	A.Den. (Anw)	A.Den. (Aw)	Den. R.(nw)	Den. R.(w)
wml	5	74	227	227	1372	2158	2080	0.9131	0.5221	0.3591	7.2	5
mahindas	71	365	1069	1069	2104	4097	3358	0.975	0.3656	0.2741	38.5	28.8
orani678	26	572	2372	2372	45919	75074	60826	0.9642	0.5608	0.2775	20.7	10.2
psmig1	260	2943	3140	3140	110032	117412	111717	0.9056	0.9039	0.8276	10.8	9.9
poli-large	206	642	7678	7675	683	8210	8186	0.9917	0.1882	0.1874	1306.9	1301.2

weighted settings, singleton nodes are subsequently assigned to the most appropriate components according to the corresponding interaction-based criterion.

The structural impact of the proposed assignment strategies is summarized in Table 9. The table follows the same structure as Table 7, including data for both unweighted and weighted variants. As expected, the inclusion of singleton nodes leads to a substantial increase in the total number of nodes and edges across all networks, indicating that the identified dense components capture only a core subset of the overall structure.

This expansion is accompanied by a decrease in the average density of components. However, the reduction remains controlled, with average densities in the unweighted setting typically decreasing from values close to 0.9–0.99 before expansion to values ranging between approximately 0.19 and 0.90 after expansion, while in the weighted setting they further decrease to a range of approximately 0.19–0.83. Despite this reduction, the resulting densities remain significantly higher than the global network density, as confirmed by the large density ratio values, which range from approximately 7 to over 1306 in the unweighted case and from approximately 5 to over 1301 in the weighted case. This demonstrates that the extended components retain a high level of internal cohesion despite their increased size.

A comparison between the unweighted and weighted variants reveals that incorporating edge weights generally leads to slightly lower densities, reflecting a more selective integration of singleton nodes based on interaction strength. In particular, the weighted approach tends to produce more conservative expansions, preserving stronger internal connectivity at the cost of reduced overall density. Notably, in networks such as *psmig1*, the density remains remarkably stable even after expansion, indicating a highly cohesive underlying structure.

All the findings presented in this section clearly indicate that the proposed approach provides a consistent and coherent framework for structural analysis across different types of networks. In particular, it is shown that the same underlying mechanism yields meaningful and interpretable component structures beyond its primary aim of GO term prediction, as clearly demonstrated on examples from both collaborative and economic settings.

6 Additional evaluation of HCDAM method

In this section, we provide a detailed overview of the external temporal validation results for the *H. sapiens* dataset. The tables 10, 11 and 12 list the Recall, Precision, and F1 scores obtained for different combinations of DINGO and STRING cutoff parameters across the three GO subontologies, BPO, MFO, and CCO, respectively. These results complement the summary presented in the main text and allow for a more comprehensive assessment of the performance of HCDAM under varying parameter settings.

Each table presents the following columns, in order: DINGO cutoff, STRING cutoff, True Positives (TP), False Positives (FP), False Negatives (FN), Recall, Precision, and F1 score. In each table, the cutoff parameter combination that was reported as optimal in the main text is highlighted in bold.

DINGO cutoff	STRNG cutoff	TP	FP	FN	Recall	Precision	F1
1.00E-05	0.222777778	20280	156160	111676	0.153688	0.11494	0.131519
1.00E-05	0.333666667	16740	109181	115216	0.12686	0.13294	0.129829
1.00E-05	0.111888889	23215	209900	108741	0.17593	0.099586	0.127181
1.00E-05	0.444555556	13620	75455	118336	0.103216	0.152905	0.123241
1.00E-05	0.001	23594	231112	108362	0.178802	0.092632	0.122039
0.005564444	0.555444444	17474	140089	114482	0.132423	0.110902	0.120711
0.005564444	0.444555556	20583	191567	111373	0.155984	0.097021	0.119632
0.011118889	0.555444444	18938	168961	113018	0.143518	0.100788	0.118416
0.005564444	0.333666667	24155	259011	107801	0.183053	0.085303	0.116375
0.011118889	0.444555556	22139	229161	109817	0.167776	0.088098	0.115531
0.016673333	0.555444444	20184	199253	111772	0.15296	0.091981	0.11488
0.011118889	0.666333333	15535	123649	116421	0.117729	0.111615	0.11459
0.005564444	0.666333333	14153	102044	117803	0.107255	0.121802	0.114067
1.00E-05	0.555444444	11189	53077	120767	0.084793	0.174105	0.114044
0.016673333	0.666333333	16777	148322	115179	0.127141	0.101618	0.112956
0.022227778	0.555444444	21287	226425	110669	0.161319	0.085934	0.112135
0.016673333	0.444555556	23580	266463	108376	0.178696	0.081298	0.111754
0.011118889	0.333666667	25829	306503	106127	0.195739	0.07772	0.111263
0.027782222	0.555444444	22115	243508	109841	0.167594	0.083257	0.111248
0.022227778	0.666333333	17793	172092	114163	0.13484	0.093704	0.11057
0.027782222	0.666333333	18565	185367	113391	0.140691	0.091035	0.110543
0.022227778	0.444555556	24843	298862	107113	0.188267	0.076746	0.109042
0.005564444	0.222777778	28388	362130	103568	0.215132	0.072693	0.108668
0.033336667	0.555444444	23160	271501	108796	0.175513	0.078599	0.108575
0.027782222	0.444555556	25851	320616	106105	0.195906	0.074613	0.108068
0.033336667	0.666333333	19533	210330	112423	0.148027	0.084977	0.107971
0.016673333	0.777222222	14229	117432	117727	0.107831	0.108073	0.107952
0.016673333	0.333666667	27673	355485	104283	0.209714	0.072223	0.107444
0.011118889	0.777222222	12916	95750	119040	0.097881	0.11886	0.107355
0.027782222	0.777222222	15979	150885	115977	0.121093	0.095761	0.106947
0.038891111	0.555444444	23928	292317	108028	0.181333	0.075663	0.106774
0.038891111	0.666333333	20269	227605	111687	0.153604	0.081771	0.106727
0.022227778	0.777222222	15214	139430	116742	0.115296	0.098381	0.106169
0.044445556	0.555444444	24609	311688	107347	0.186494	0.073176	0.10511
0.033336667	0.444555556	26878	353112	105078	0.203689	0.070733	0.105003
0.044445556	0.666333333	20806	244342	111150	0.157674	0.078469	0.104789
0.033336667	0.777222222	16881	173574	115075	0.127929	0.088635	0.104717
0.022227778	0.333666667	29017	395316	102939	0.219899	0.068383	0.104323
0.038891111	0.777222222	17568	188827	114388	0.133135	0.085118	0.103845
0.05	0.555444444	25111	328469	106845	0.190298	0.071019	0.103436
0.011118889	0.222777778	30434	426241	101522	0.230637	0.066643	0.103406
0.05	0.666333333	21291	258968	110665	0.161349	0.075969	0.1033

0.027782222	0.333666667	30164	423867	101792	0.228591	0.066436	0.102951
0.005564444	0.777222222	11448	79052	120508	0.086756	0.126497	0.102924
0.038891111	0.444555556	27654	378365	104302	0.20957	0.06811	0.102808
0.044445556	0.777222222	18113	204161	113843	0.137265	0.08149	0.102267
0.05	0.777222222	18627	217332	113329	0.141161	0.078942	0.101257
0.044445556	0.444555556	28417	401652	103539	0.215352	0.066075	0.101124
0.027782222	0.888111111	13487	121969	118469	0.102208	0.099567	0.100871
0.033336667	0.333666667	31384	463650	100572	0.237837	0.063398	0.10011
0.033336667	0.888111111	14391	141365	117565	0.109059	0.092395	0.100038
0.038891111	0.888111111	15026	154236	116930	0.113871	0.088774	0.099768
0.016673333	0.222777778	32731	492121	99225	0.248045	0.062362	0.099667
0.05	0.444555556	29057	422186	102899	0.220202	0.064393	0.099647
0.022227778	0.888111111	12775	112214	119181	0.096813	0.102209	0.099438
0.005564444	0.111888889	32143	483448	99813	0.243589	0.062342	0.099276
1.00E-05	0.666333333	8837	38646	123119	0.066969	0.186109	0.098496
0.044445556	0.888111111	15535	168238	116421	0.117729	0.084534	0.098407
0.016673333	0.888111111	11659	93406	120297	0.088355	0.110969	0.098379
0.038891111	0.333666667	32412	496476	99544	0.245627	0.061283	0.098093
0.05	0.888111111	16008	179006	115948	0.121313	0.082086	0.097917
0.022227778	0.222777778	34400	547170	97556	0.260693	0.05915	0.096423
0.044445556	0.333666667	33231	525886	98725	0.251834	0.059435	0.096172
0.011118889	0.888111111	10477	76547	121479	0.079398	0.120392	0.095689
0.027782222	0.222777778	35868	589303	96088	0.271818	0.057373	0.094748
0.011118889	0.111888889	34830	571279	97126	0.263952	0.057465	0.094382
0.05	0.333666667	33847	552185	98109	0.256502	0.057756	0.094283
0.005564444	0.001	32655	534766	99301	0.247469	0.05755	0.093383
0.033336667	0.222777778	37317	642553	94639	0.282799	0.054888	0.091933
0.005564444	0.888111111	9196	63110	122760	0.06969	0.127182	0.090041
0.016673333	0.111888889	37366	662228	94590	0.28317	0.053411	0.089871
0.038891111	0.222777778	38458	686586	93498	0.291446	0.053042	0.08975
0.011118889	0.001	35299	629710	96657	0.267506	0.05308	0.088584
0.044445556	0.222777778	39375	727668	92581	0.298395	0.051333	0.087597
0.022227778	0.111888889	39323	735021	92633	0.298001	0.050782	0.086777
0.05	0.222777778	40095	765481	91861	0.303851	0.049772	0.085533
0.027782222	0.111888889	41171	798814	90785	0.312006	0.049014	0.084719
0.016673333	0.001	37823	726750	94133	0.286633	0.049469	0.084377
0.033336667	0.111888889	42793	868291	89163	0.324297	0.046969	0.082054
0.022227778	0.001	39762	801994	92194	0.301328	0.047237	0.081671
1.00E-05	0.777222222	6914	30914	125042	0.052396	0.182775	0.081445
0.038891111	0.111888889	44298	927674	87658	0.335703	0.045575	0.080255
0.027782222	0.001	41617	870302	90339	0.315385	0.045637	0.079736
0.044445556	0.111888889	45392	983046	86564	0.343993	0.044137	0.078235
0.033336667	0.001	43238	941672	88718	0.32767	0.0439	0.077427
0.05	0.111888889	46339	1035462	85617	0.35117	0.042835	0.076356

0.038891111	0.001	44726	1003382	87230	0.338946	0.042673	0.075803
0.044445556	0.001	45796	1060986	86160	0.347055	0.041378	0.07394
0.05	0.001	46721	1116852	85235	0.354065	0.040153	0.072127
1.00E-05	0.888111111	5038	24066	126918	0.038179	0.173103	0.062561
0.05	0.999	2153	11485	129803	0.016316	0.157868	0.029575
0.044445556	0.999	2058	10245	129898	0.015596	0.167276	0.028532
0.038891111	0.999	1842	7807	130114	0.013959	0.190901	0.026016
0.033336667	0.999	1705	6475	130251	0.012921	0.208435	0.024334
0.027782222	0.999	1425	4707	130531	0.010799	0.232387	0.020639
0.022227778	0.999	1219	3721	130737	0.009238	0.246761	0.017809
0.016673333	0.999	975	2636	130981	0.007389	0.270008	0.014384
0.011118889	0.999	756	1617	131200	0.005729	0.318584	0.011256
0.005564444	0.999	510	912	131446	0.003865	0.35865	0.007647
1.00E-05	0.999	82	38	131874	6.21E-04	0.683333	0.001242

Table 10: Detailed external validation results for H. sapiens – Biological Process Ontology

DINGO cutoff	STRNG cutoff	TP	FP	FN	Recall	Precision	F1
1.00E-05	0.222777778	4381	23766	26549	0.141642	0.155647	0.148315
0.005564444	0.333666667	5028	34347	25902	0.162561	0.127695	0.143034
1.00E-05	0.111888889	5268	37892	25662	0.17032	0.122057	0.142205
0.011118889	0.333666667	5484	40899	25446	0.177304	0.118233	0.141865
0.005564444	0.222777778	6260	51381	24670	0.202392	0.108603	0.141356
0.011118889	0.444555556	4454	28182	26476	0.144003	0.136475	0.140138
0.016673333	0.333666667	5785	46660	25145	0.187035	0.110306	0.138771
0.016673333	0.444555556	4704	32566	26226	0.152085	0.126214	0.137947
0.011118889	0.222777778	6752	61419	24178	0.218299	0.099045	0.136265
0.022227778	0.444555556	4851	35624	26079	0.156838	0.119852	0.135873
0.022227778	0.333666667	5939	50744	24991	0.192014	0.104776	0.135573
0.027782222	0.444555556	5062	38702	25868	0.16366	0.115666	0.13554
0.005564444	0.444555556	3928	23187	27002	0.126996	0.144864	0.135343
0.033336667	0.444555556	5163	40259	25767	0.166925	0.113667	0.135242
1.00E-05	0.333666667	3316	14900	27614	0.10721	0.182038	0.134945
1.00E-05	0.001	5450	44625	25480	0.176204	0.108837	0.13456
0.027782222	0.333666667	6138	54692	24792	0.198448	0.100904	0.133784
0.033336667	0.333666667	6261	56850	24669	0.202425	0.099206	0.133155
0.038891111	0.444555556	5251	43228	25679	0.16977	0.108315	0.132252
0.016673333	0.222777778	7074	69757	23856	0.22871	0.092072	0.131291
0.044445556	0.444555556	5311	45194	25619	0.17171	0.105158	0.130435
0.05	0.444555556	5413	47160	25517	0.175008	0.102962	0.129648
0.038891111	0.333666667	6343	60730	24587	0.205076	0.094569	0.129445
0.022227778	0.222777778	7262	75411	23668	0.234788	0.08784	0.127849
0.033336667	0.555444444	4116	29343	26814	0.133075	0.123016	0.127848
0.044445556	0.333666667	6419	63444	24511	0.207533	0.09188	0.12737
0.027782222	0.555444444	4015	28149	26915	0.129809	0.124829	0.12727
0.011118889	0.555444444	3436	19684	27494	0.11109	0.148616	0.127142
0.016673333	0.555444444	3669	23271	27261	0.118623	0.136192	0.126801
0.022227778	0.555444444	3823	25649	27107	0.123602	0.129716	0.126585
0.038891111	0.555444444	4209	31653	26721	0.136081	0.117367	0.126033
0.05	0.333666667	6511	66003	24419	0.210508	0.08979	0.125885
0.005564444	0.111888889	7172	75933	23758	0.231878	0.0863	0.125786
0.027782222	0.222777778	7489	80870	23441	0.242127	0.084757	0.125561
0.05	0.555444444	4386	34578	26544	0.141804	0.112565	0.125504
0.044445556	0.555444444	4279	33132	26651	0.138345	0.114378	0.125225
0.033336667	0.222777778	7624	83739	23306	0.246492	0.083447	0.124684
0.038891111	0.222777778	7721	88854	23209	0.249628	0.079948	0.121109
0.011118889	0.111888889	7749	90602	23181	0.250533	0.078789	0.119878
0.044445556	0.222777778	7822	92686	23108	0.252894	0.077825	0.119022
0.005564444	0.555444444	2965	15968	27965	0.095862	0.156605	0.118926
0.005564444	0.001	7331	86383	23599	0.237019	0.078227	0.117631

0.05	0.222777778	7924	96382	23006	0.256191	0.075969	0.117188
0.05	0.666333333	3494	26371	27436	0.112965	0.116993	0.114944
0.016673333	0.111888889	8143	102856	22787	0.263272	0.073361	0.114748
0.033336667	0.666333333	3228	22201	27702	0.104365	0.126942	0.114551
0.038891111	0.666333333	3323	24077	27607	0.107436	0.121277	0.113938
0.044445556	0.666333333	3393	25253	27537	0.109699	0.118446	0.113905
0.027782222	0.666333333	3135	21213	27795	0.101358	0.128758	0.113427
1.00E-05	0.444555556	2428	9526	28502	0.0785	0.203112	0.113236
0.011118889	0.001	7909	101440	23021	0.255706	0.072328	0.112761
0.022227778	0.111888889	8429	111472	22501	0.272519	0.0703	0.111767
0.022227778	0.666333333	2953	19139	27977	0.095474	0.133668	0.111388
0.016673333	0.666333333	2817	17209	28113	0.091077	0.140667	0.110566
0.027782222	0.111888889	8709	118849	22221	0.281571	0.068275	0.109901
0.033336667	0.111888889	8899	122788	22031	0.287714	0.067577	0.109447
0.011118889	0.666333333	2610	14295	28320	0.084384	0.154392	0.109125
0.016673333	0.001	8296	114657	22634	0.268219	0.067473	0.107822
0.038891111	0.111888889	9077	130421	21853	0.293469	0.065069	0.10652
0.022227778	0.001	8574	123528	22356	0.277207	0.064904	0.105182
0.05	0.777222222	2892	21201	28038	0.093501	0.120035	0.10512
0.044445556	0.111888889	9220	135964	21710	0.298092	0.063506	0.104705
0.033336667	0.777222222	2669	17779	28261	0.086292	0.130526	0.103897
0.044445556	0.777222222	2805	20268	28125	0.090689	0.121571	0.103883
0.027782222	0.001	8851	131078	22079	0.286162	0.063254	0.103606
0.038891111	0.777222222	2742	19353	28188	0.088652	0.1241	0.103423
0.05	0.111888889	9364	140931	21566	0.302748	0.062304	0.103341
0.033336667	0.001	9043	135103	21887	0.29237	0.062735	0.103304
0.005564444	0.666333333	2248	11320	28682	0.07268	0.165684	0.101038
0.038891111	0.001	9214	142792	21716	0.297898	0.060616	0.100735
0.027782222	0.777222222	2538	16923	28392	0.082056	0.130415	0.100732
0.022227778	0.777222222	2414	15178	28516	0.078047	0.137221	0.099501
0.044445556	0.001	9360	148394	21570	0.302619	0.059333	0.099213
0.05	0.001	9501	153423	21429	0.307177	0.058316	0.098022
0.016673333	0.777222222	2284	13556	28646	0.073844	0.144192	0.097669
0.05	0.888111111	2441	16945	28489	0.07892	0.125916	0.097027
0.044445556	0.888111111	2371	16178	28559	0.076657	0.127824	0.095839
0.033336667	0.888111111	2262	14174	28668	0.073133	0.137625	0.095512
0.038891111	0.888111111	2321	15421	28609	0.07504	0.13082	0.095373
0.011118889	0.777222222	2047	11178	28883	0.066182	0.154783	0.092719
1.00E-05	0.555444444	1794	5995	29136	0.058002	0.230325	0.092668
0.027782222	0.888111111	2128	13466	28802	0.068801	0.136463	0.09148
0.022227778	0.888111111	2019	11918	28911	0.065276	0.144866	0.089999
0.016673333	0.888111111	1900	10609	29030	0.061429	0.151891	0.087479
0.005564444	0.777222222	1747	8587	29183	0.056482	0.169054	0.084674
0.011118889	0.888111111	1710	8611	29220	0.055286	0.165682	0.082907

1.00E-05	0.666333333	1382	4133	29548	0.044682	0.250589	0.07584
0.005564444	0.888111111	1452	6569	29478	0.046945	0.181025	0.074555
1.00E-05	0.777222222	1049	3054	29881	0.033915	0.255667	0.059886
1.00E-05	0.888111111	827	2280	30103	0.026738	0.266173	0.048594
0.05	0.999	220	927	30710	0.007113	0.191805	0.013717
0.044445556	0.999	202	854	30728	0.006531	0.191288	0.012631
0.038891111	0.999	199	805	30731	0.006434	0.198207	0.012463
0.033336667	0.999	178	730	30752	0.005755	0.196035	0.011182
0.027782222	0.999	167	659	30763	0.005399	0.202179	0.010518
0.022227778	0.999	118	418	30812	0.003815	0.220149	0.0075
0.016673333	0.999	82	323	30848	0.002651	0.202469	0.005234
0.011118889	0.999	45	255	30885	0.001455	0.15	0.002882
0.005564444	0.999	38	214	30892	0.001229	0.150794	0.002437
1.00E-05	0.999	2	4	30928	6.47E-05	0.333333	1.29E-04

Table 11: Detailed external validation results for H. sapiens – Molecular Function Ontology

DINGO cutoff	STRNG cutoff	TP	FP	FN	Recall	Precision	F1
0.016673	0.111889	15032	66904	40824	0.269121	0.18346	0.218184
1.00E-05	0.001	10116	27763	45740	0.181109	0.267061	0.215843
0.011119	0.111889	14109	60771	41747	0.252596	0.188421	0.21584
0.016673	0.222778	12910	50888	42946	0.23113	0.202357	0.215789
0.022228	0.111889	15636	73455	40220	0.279934	0.175506	0.215748
0.038891	0.222778	14677	65907	41179	0.262765	0.182133	0.215142
0.022228	0.222778	13476	56123	42380	0.241263	0.193623	0.214834
0.011119	0.222778	12298	46439	43558	0.220173	0.209374	0.214638
0.033337	0.222778	14194	62264	41662	0.254118	0.185644	0.21455
0.027782	0.111889	16100	78132	39756	0.288241	0.170855	0.214541
0.027782	0.222778	13866	59630	41990	0.248245	0.188663	0.214392
1.00E-05	0.111889	9760	25475	46096	0.174735	0.276997	0.214291
0.033337	0.111889	16481	81655	39375	0.295062	0.16794	0.21405
0.044446	0.222778	14930	68764	40926	0.267294	0.178388	0.213973
0.05	0.222778	15212	71379	40644	0.272343	0.175676	0.213581
0.038891	0.111889	16990	86408	38866	0.304175	0.164317	0.21337
0.005564	0.111889	12935	52594	42921	0.231578	0.197394	0.213124
0.016673	0.001	15318	73214	40538	0.274241	0.173022	0.212178
0.044446	0.111889	17343	90340	38513	0.310495	0.161056	0.212096
0.038891	0.333667	12449	49238	43407	0.222877	0.201809	0.21182
0.033337	0.333667	12132	46657	43724	0.217201	0.206365	0.211645
0.005564	0.222778	11362	40205	44494	0.203416	0.220335	0.211538
0.027782	0.333667	11882	44748	43974	0.212726	0.209818	0.211262
0.05	0.111889	17660	93907	38196	0.31617	0.158291	0.210963
0.05	0.333667	12872	53380	42984	0.23045	0.194288	0.21083
0.044446	0.333667	12623	51323	43233	0.225992	0.197401	0.210731
0.016673	0.333667	11100	38425	44756	0.198725	0.224129	0.210664
0.022228	0.333667	11555	42336	44301	0.206871	0.214414	0.210575
0.011119	0.001	14405	66787	41451	0.257895	0.177419	0.210218
0.022228	0.001	15907	80055	39949	0.284786	0.165764	0.209554
0.027782	0.001	16382	85019	39474	0.29329	0.161557	0.208347
0.005564	0.001	13224	58067	42632	0.236752	0.185493	0.208011
0.033337	0.001	16774	88695	39082	0.300308	0.159042	0.207953
0.011119	0.333667	10540	34990	45316	0.1887	0.231496	0.207918
0.038891	0.001	17261	93537	38595	0.309027	0.155788	0.207148
0.044446	0.001	17603	97528	38253	0.31515	0.152895	0.205899
0.038891	0.444556	10736	38080	45120	0.192209	0.219928	0.205136
0.05	0.001	17919	101174	37937	0.320807	0.150462	0.204848
0.033337	0.444556	10478	36155	45378	0.18759	0.224691	0.204471
0.005564	0.333667	9826	30575	46030	0.175917	0.243212	0.204162
0.044446	0.444556	10839	39637	45017	0.194053	0.214736	0.203871
0.027782	0.444556	10261	34584	45595	0.183705	0.22881	0.203791

0.05	0.444556	11009	41210	44847	0.197096	0.210824	0.203729
0.022228	0.444556	9933	32591	45923	0.177832	0.233586	0.201931
1.00E-05	0.222778	8514	20107	47342	0.152428	0.297474	0.20157
0.016673	0.444556	9544	29507	46312	0.170868	0.244398	0.201123
0.011119	0.444556	9017	26736	46839	0.161433	0.252203	0.196858
0.038891	0.555444	9252	29673	46604	0.16564	0.237688	0.195229
0.05	0.555444	9512	32183	46344	0.170295	0.228133	0.195016
0.044446	0.555444	9377	30946	46479	0.167878	0.232547	0.194991
0.033337	0.555444	9000	27965	46856	0.161129	0.243474	0.193922
0.027782	0.555444	8839	26631	47017	0.158246	0.249197	0.19357
0.022228	0.555444	8582	25049	47274	0.153645	0.255181	0.191804
0.005564	0.444556	8351	23223	47505	0.149509	0.26449	0.191033
0.016673	0.555444	8195	22599	47661	0.146717	0.266123	0.189152
0.05	0.666333	8400	25982	47456	0.150387	0.244314	0.186174
0.038891	0.666333	8127	23842	47729	0.145499	0.254215	0.185073
0.044446	0.666333	8229	24912	47627	0.147325	0.248303	0.184928
0.011119	0.555444	7773	20493	48083	0.139161	0.274995	0.184803
0.033337	0.666333	7906	22398	47950	0.141543	0.26089	0.183519
1.00E-05	0.333667	7211	15739	48645	0.1291	0.314205	0.183006
0.027782	0.666333	7734	21202	48122	0.138463	0.26728	0.182423
0.022228	0.666333	7525	19936	48331	0.134721	0.274025	0.180635
0.005564	0.555444	7216	17994	48640	0.129189	0.286236	0.178028
0.016673	0.666333	7192	17876	48664	0.12876	0.2869	0.177747
0.011119	0.666333	6881	16445	48975	0.123192	0.294993	0.173802
0.05	0.777222	7243	20844	48613	0.129673	0.257877	0.172569
0.038891	0.777222	7008	18943	48848	0.125465	0.270047	0.17133
0.044446	0.777222	7072	19926	48784	0.126611	0.261945	0.17071
0.033337	0.777222	6808	17609	49048	0.121885	0.278822	0.169621
0.027782	0.777222	6664	16633	49192	0.119307	0.286045	0.168383
0.022228	0.777222	6498	15609	49358	0.116335	0.293934	0.166694
0.005564	0.666333	6374	14506	49482	0.114115	0.305268	0.166128
1.00E-05	0.444556	6105	12125	49751	0.109299	0.334888	0.164808
0.016673	0.777222	6165	14134	49691	0.110373	0.30371	0.161907
0.011119	0.777222	5857	12893	49999	0.104859	0.312373	0.157012
0.05	0.888111	6213	17100	49643	0.111232	0.266504	0.156955
0.038891	0.888111	6019	15378	49837	0.107759	0.281301	0.155826
0.044446	0.888111	6080	16311	49776	0.108851	0.271538	0.155405
0.033337	0.888111	5848	14123	50008	0.104698	0.292825	0.154246
0.027782	0.888111	5696	13243	50160	0.101977	0.300755	0.15231
1.00E-05	0.555444	5402	9874	50454	0.096713	0.353627	0.151887
0.022228	0.888111	5548	12389	50308	0.099327	0.309305	0.150367
0.005564	0.777222	5396	11285	50460	0.096606	0.323482	0.148779
0.016673	0.888111	5194	11175	50662	0.092989	0.317307	0.143828
0.011119	0.888111	4969	10194	50887	0.088961	0.327706	0.139934

1.00E-05	0.666333	4787	8065	51069	0.085703	0.372471	0.139343
0.005564	0.888111	4529	8819	51327	0.081084	0.339302	0.130888
1.00E-05	0.777222	3956	6216	51900	0.070825	0.388911	0.119828
1.00E-05	0.888111	3239	4615	52617	0.057988	0.412401	0.101679
0.05	0.999	1360	1455	54496	0.024348	0.483126	0.04636
0.044446	0.999	1322	1364	54534	0.023668	0.492182	0.045164
0.038891	0.999	1283	1295	54573	0.02297	0.497673	0.043913
0.033337	0.999	1214	1201	54642	0.021734	0.502692	0.041667
0.027782	0.999	1106	1094	54750	0.019801	0.502727	0.038101
0.022228	0.999	1033	967	54823	0.018494	0.5165	0.035709
0.016673	0.999	680	732	55176	0.012174	0.481586	0.023748
0.011119	0.999	583	584	55273	0.010438	0.499572	0.020448
0.005564	0.999	394	336	55462	0.007054	0.539726	0.013926
1.00E-05	0.999	7	11	55849	1.25E-04	0.388889	2.51E-04

Table 12: Detailed external validation results for H. sapiens – Cellular Component Ontology

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